FREIGHT RATE PREDICTION

Machine Learning Engineer Assessment Report

Spotter

Final Solution: CatBoost + Route + Date Features

1. Objective

The objective of this assessment was to develop a machine learning model capable of predicting freight posted rates from historical shipment data. The labeled development dataset contains 48,000 loads covering January through October 2025, while the final validation dataset contains 12,000 unlabeled loads covering November and December 2025.

The solution focused on:

- exploring the available freight data and identifying data-quality issues;
- selecting a time-aware validation strategy;
- comparing multiple regression approaches;
- engineering directional route and calendar features;
- training the selected model on all labeled development data; and
- generating the required validation and fixed December predictions.

2. Data Exploration

Exploratory data analysis examined feature distributions, temporal behavior, data quality, and relationships between the available predictors and posted_rate.

Key Findings

Distance was the strongest individual predictor. Its relationship with posted_rate was approximately 0.91 by Pearson correlation and 0.98 by Spearman correlation, indicating a strong, mostly monotonic relationship between shipment distance and freight rate.

A directional route feature was constructed as pickup -> delivery. Direction was preserved because freight-market behavior can differ by direction even for the same pair of locations.

The date field was also analyzed by month, day of week, day of month, and overall time trend. This motivated testing calendar-derived features during model development.

3. Data Quality and Preprocessing

Missing Values

The training data contained approximately 300 missing weight values and 374 missing market_index values. The final validation data contained approximately 165 missing weight values and 249 missing market_index values.

Missing numerical values were imputed using medians calculated from the training data only. Missing categorical values were represented as UNKNOWN. This preprocessing design avoids using information from the future validation period.

Negative Weights

Approximately 292 negative weight values were observed in training and 145 in validation. Because shipment weight should not be negative while the magnitude remains meaningful, these values were corrected using their absolute value.

Duplicates and High-Value Loads

No duplicate records requiring removal were identified. High-value posted rates were retained because they represented plausible freight observations rather than clear data-entry errors.

4. Validation Strategy

A random split was intentionally avoided because the data is chronological and the final prediction period occurs after the labeled development period. A random split could therefore produce an overly optimistic estimate of future performance.

Fold	Training period	Validation period
1	Jan-Aug 2025	Sep-Oct 2025
2	Jan-Sep 2025	Oct 2025

This temporal validation strategy better approximates the production setting in which historical freight data is used to predict future loads. After model and feature selection, the final model was trained on all labeled data from January through October 2025.

5. Model Development

Model	Internal validation MAE
Distance-only baseline	~194.86
Numerical features	~194.74
Numerical + categorical features	~162.22
CatBoost	~123.69
CatBoost + log target	~144.10
CatBoost + directional route	~123.05

Categorical information produced a substantial improvement over numerical-only baselines. CatBoost was selected because it models nonlinear relationships effectively and supports categorical variables without extensive one-hot encoding. A log-transformed target and other engineered features that weakened temporal validation performance were rejected.

6. Date Feature Experiment

The initial CatBoost + Route model did not directly use the shipment date. Calendar features were therefore evaluated using the same temporal validation framework rather than being added solely to create variation in the December chart.

- month
- day_of_week
- day

Adding these features improved average internal temporal validation MAE from approximately 122.79 to 110.88, an improvement of about 9.7%. A continuous date-ordinal feature was also tested but performed worse and was not retained.

7. Final Model

The final solution uses CatBoostRegressor with directional route and date features.

Numerical Features

- distance, weight

- pickup_lat, pickup_lon, delivery_lat, delivery_lon
- market_index, quote_signal
- month, day_of_week, day

Categorical Features

- pickup
- delivery
- equipment
- route

Model Configuration

Parameter	Value
Loss function	RMSE
Iterations	165
Learning rate	0.05
Depth	7
L2 leaf regularization	5
Random seed	42

8. Feature Importance

Feature	Importance
Distance	90.22%
Equipment	3.74%
Weight	1.20%
Month	0.95%
Market Index	0.86%
Quote Signal	0.56%
Pickup Latitude	0.46%
Day	0.42%
Delivery Latitude	0.41%
Pickup	0.29%
Delivery	0.26%
Delivery Longitude	0.22%
Route	0.15%
Pickup Longitude	0.14%
Day of Week	0.12%

Distance remained overwhelmingly dominant, consistent with the exploratory analysis. The remaining features provide smaller adjustments related to equipment, shipment weight, market conditions, geography, route, and timing.

9. Final Validation Predictions

After model selection, the final model was retrained on all 48,000 labeled development rows and used to generate predictions for every one of the 12,000 loads in validation.csv.

```
validation_predictions.csv -> load_id,predicted_rate
```

- 12,000 prediction rows
- 12,000 unique load IDs
- no missing predictions
- no duplicate IDs
- no non-positive predictions
- no non-finite predictions

The provided scorer successfully validated the output structure. The hidden final validation metrics are calculated by Spotter after submission; therefore 110.88 is reported only as the internal temporal validation MAE, not as the hidden final score.

10. December Prediction Task

The fixed December scenario contains one row for every day of December 2025 with shipment characteristics held constant: Lexington to Fort Wayne, 360 miles, Dry Van, and 32,000 lb. Only the date changes.

Handling Features Not Supplied in the December Input

The final model uses predictors not explicitly supplied in the fixed December file. Geographic coordinates were recovered from the labeled training data using city-specific medians rather than unrelated global coordinate medians. For market_index and quote_signal, route-specific training medians were used when the route had been observed historically, with global training medians as a fallback. Calendar features were generated directly from each December date.

December Results

Statistic	Predicted rate
Mean	$854.77
Minimum	$845.74
Maximum	$860.14
Standard deviation	$2.90

The narrow range is expected because shipment characteristics are intentionally fixed and only calendar features vary. These predictions should be interpreted as a fixed-route scenario, not as a full representation of daily freight-market volatility, since future market_index and quote_signal values are not supplied.

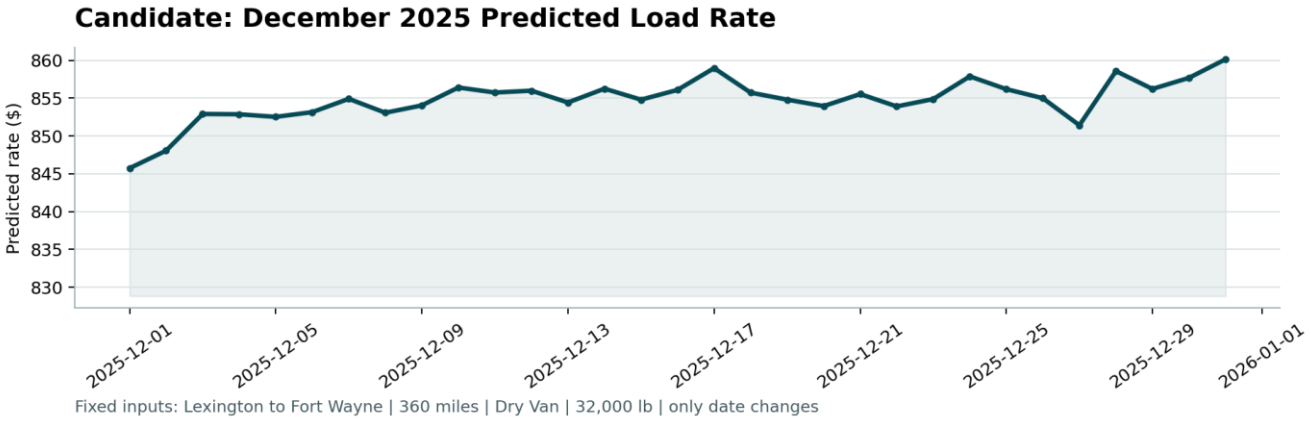


Figure 1. December 2025 predicted load rates for the fixed Lexington-to-Fort Wayne scenario.

11. Output Validation

The provided scorer was executed with:

```
python score.py --predictions validation_predictions.csv --december-predictions
data/december_chart_inputs.csv
```

The scorer returned:

```
Validated 12,000 final predictions.
Validated 31 fixed December predictions.
Created chart: scorer_results/candidate_december.png
```

This confirmed that both required prediction files satisfied the scorer's structural and validity checks.

12. Conclusion

The final solution combines exploratory analysis, leakage-aware preprocessing, temporal validation, categorical modeling, directional route information, and calendar-based feature engineering. Distance was the strongest predictor; categorical shipment information materially improved the baseline; and CatBoost provided the strongest performance among the tested models.

Calendar features improved internal temporal validation performance and provided date-related variation in the fixed December scenario. The final model achieved an average internal temporal validation MAE of approximately 110.88 before being retrained on all 48,000 labeled observations. Both the 12,000-load validation output and the 31-row December output passed the provided scorer.